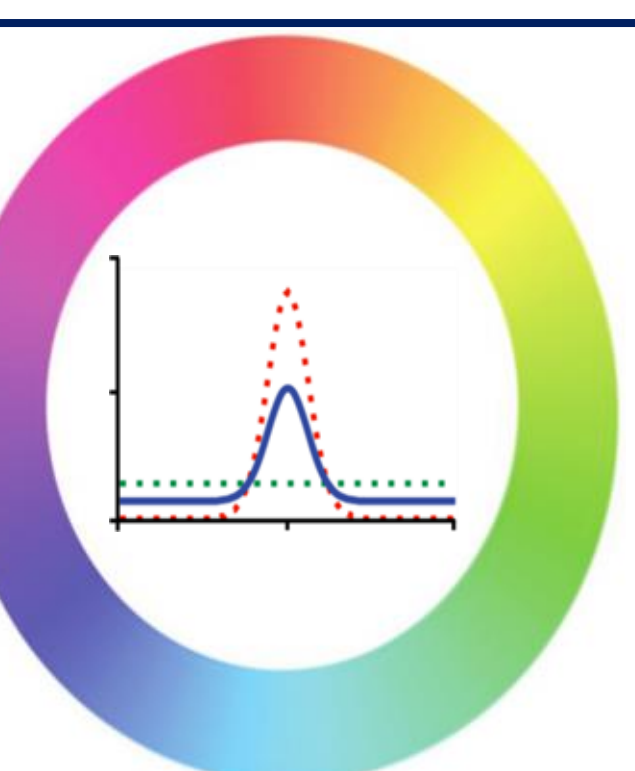




Dissociating Strength and Precision in Visual Working Memory for Motion Direction

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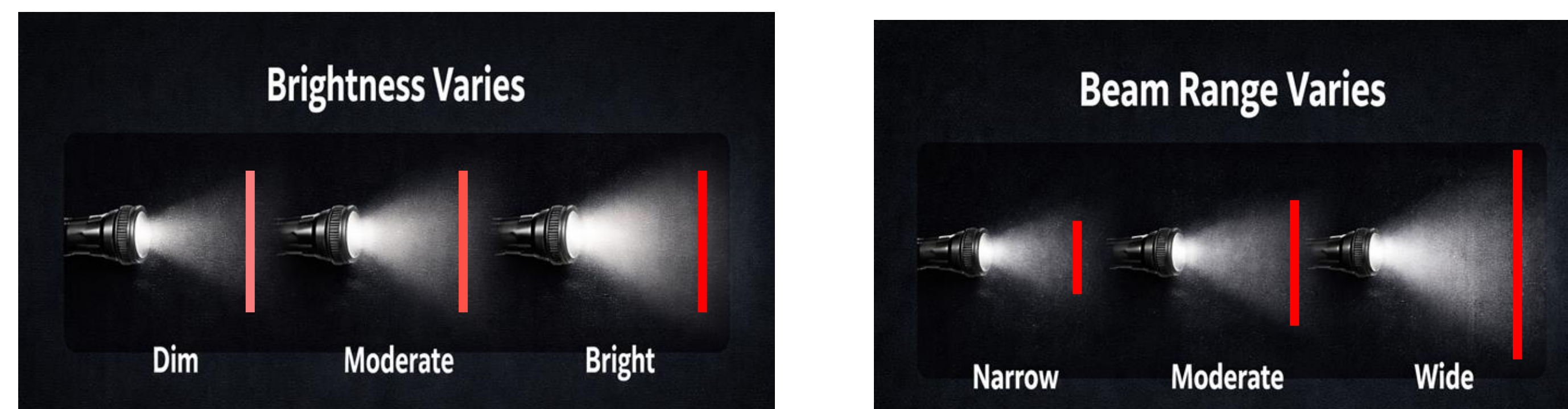
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INTRODUCTION

General Background

Flashlight beam is determined by both brightness (strength) and beam range (precision)



Likewise, noisy visual representation may be supported by both strength and precision

Visual working memory (VWM): temporary storage of a representation of visual information

- strength : distinctiveness of the representation
- precision : discriminability of the representation from similar lures

Hypothesis

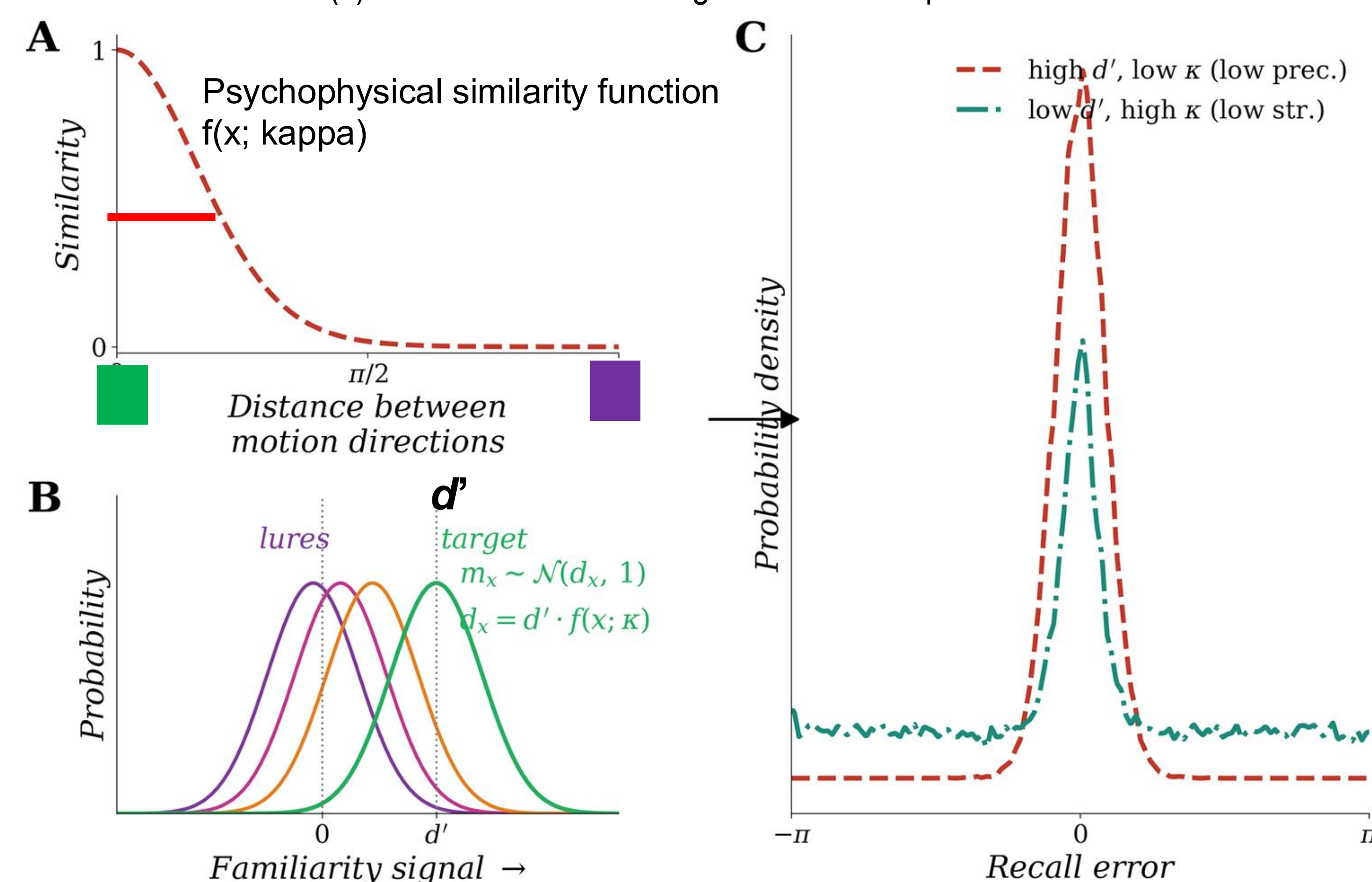
- Strength and precision are separable representational features of VWM

Target Confusability Competition (TCC)¹ + Precision model

- TCC precision explains memory recall arises from the joint contribution of strength (d') and precision (k)

Strength (d') scales the mnemonic signal of the target representation

Precision (k) determines how distinguishable the representation is from similar lures.

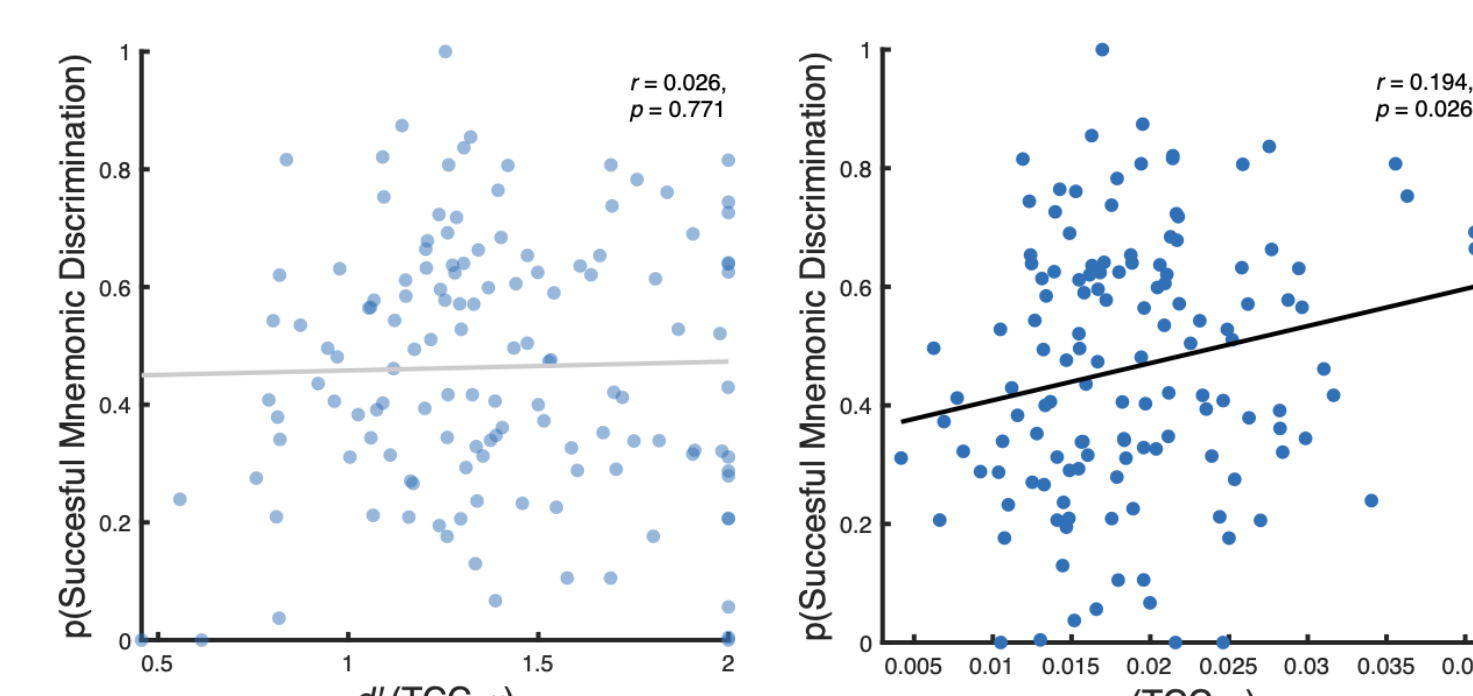


Why TCC precision?

Memory strength (d') from TCC is a compound measure, failing to capture mnemonic discrimination²

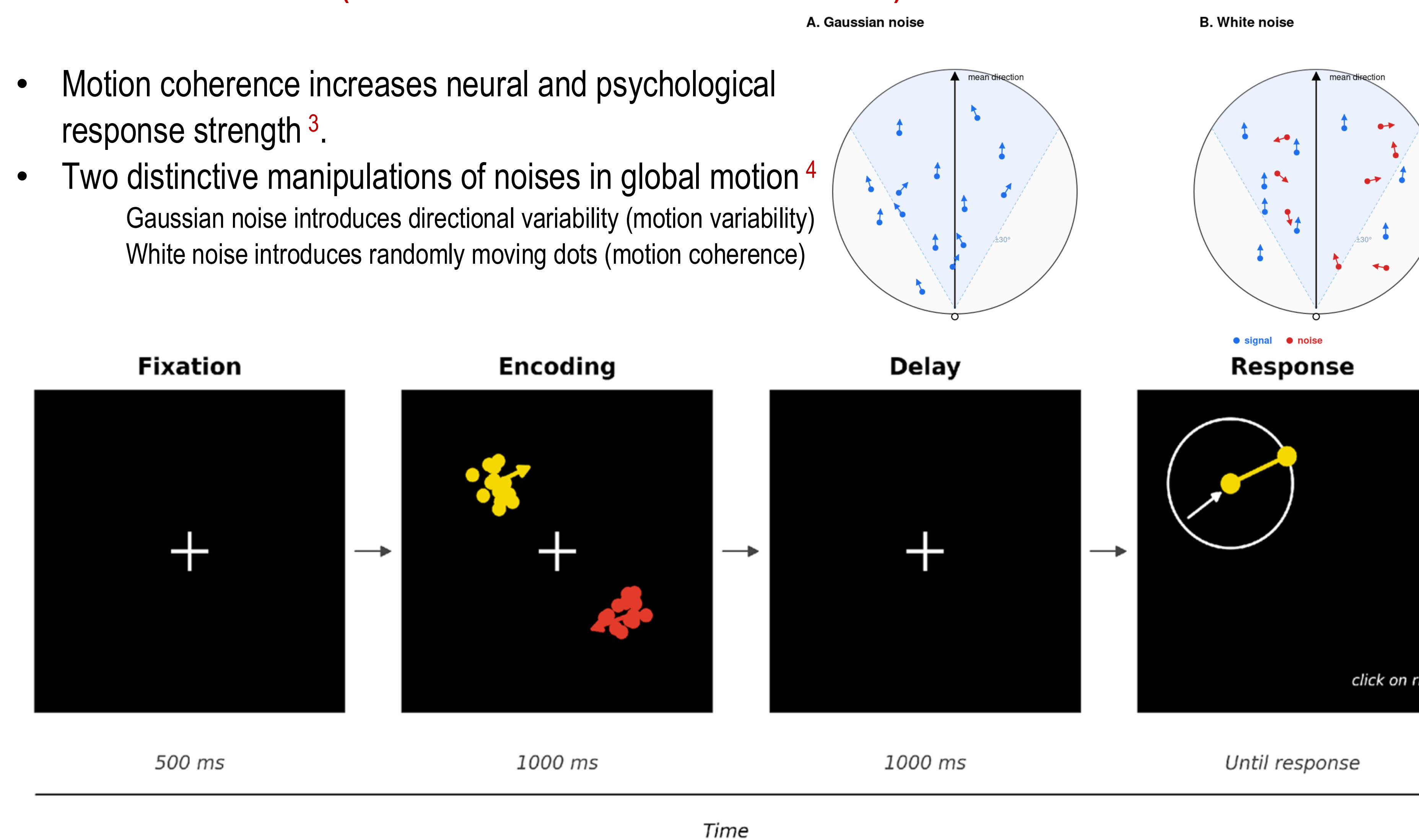
Reanalysis of Xie et al. (2024)² data

- TCC's precision (k), not strength (d') was associated with mnemonic discrimination, suggesting both estimates reflect distinct representational aspects.



METHOD AND RESULTS

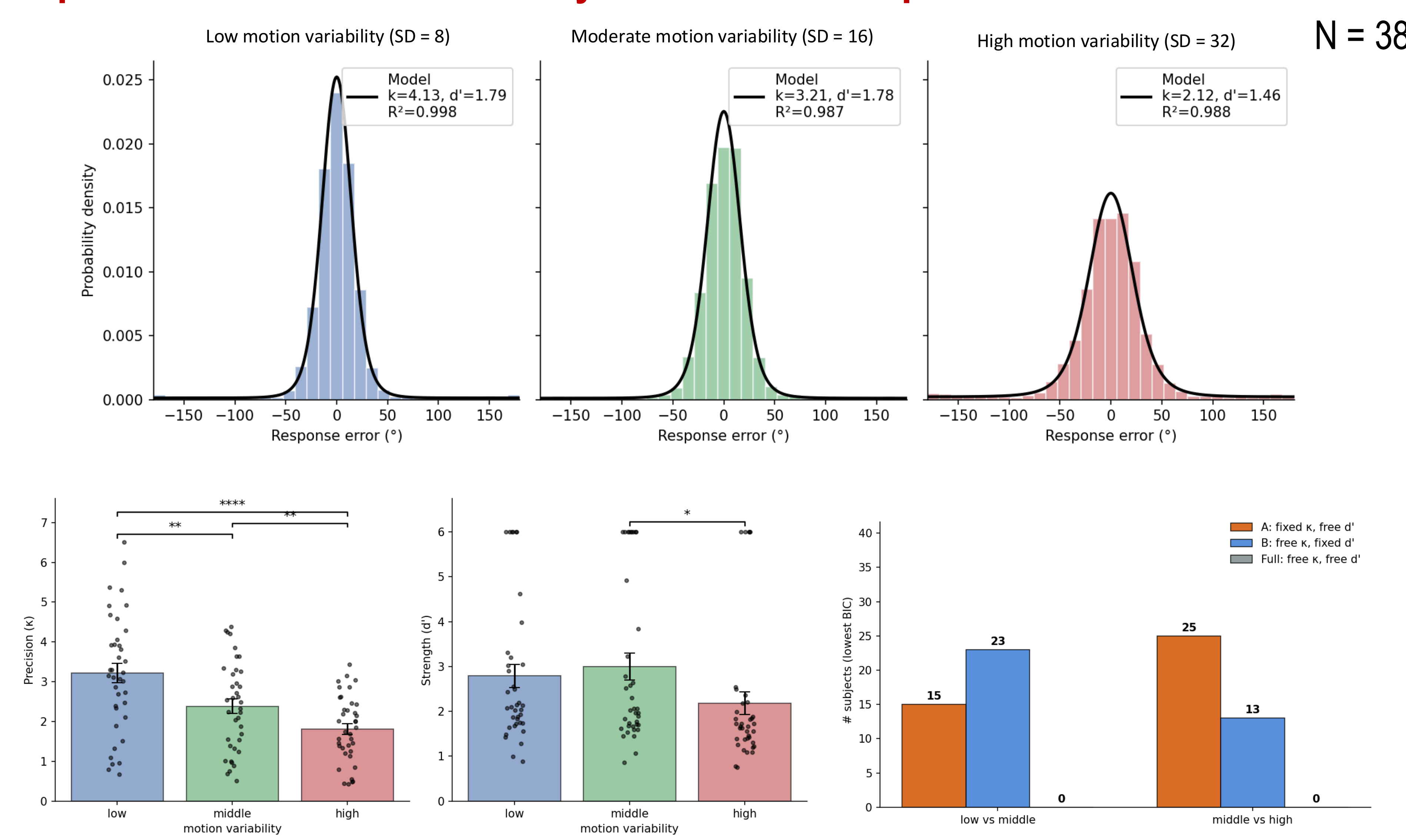
Stimuli and Task (Random dot motion recall task)



Data analysis

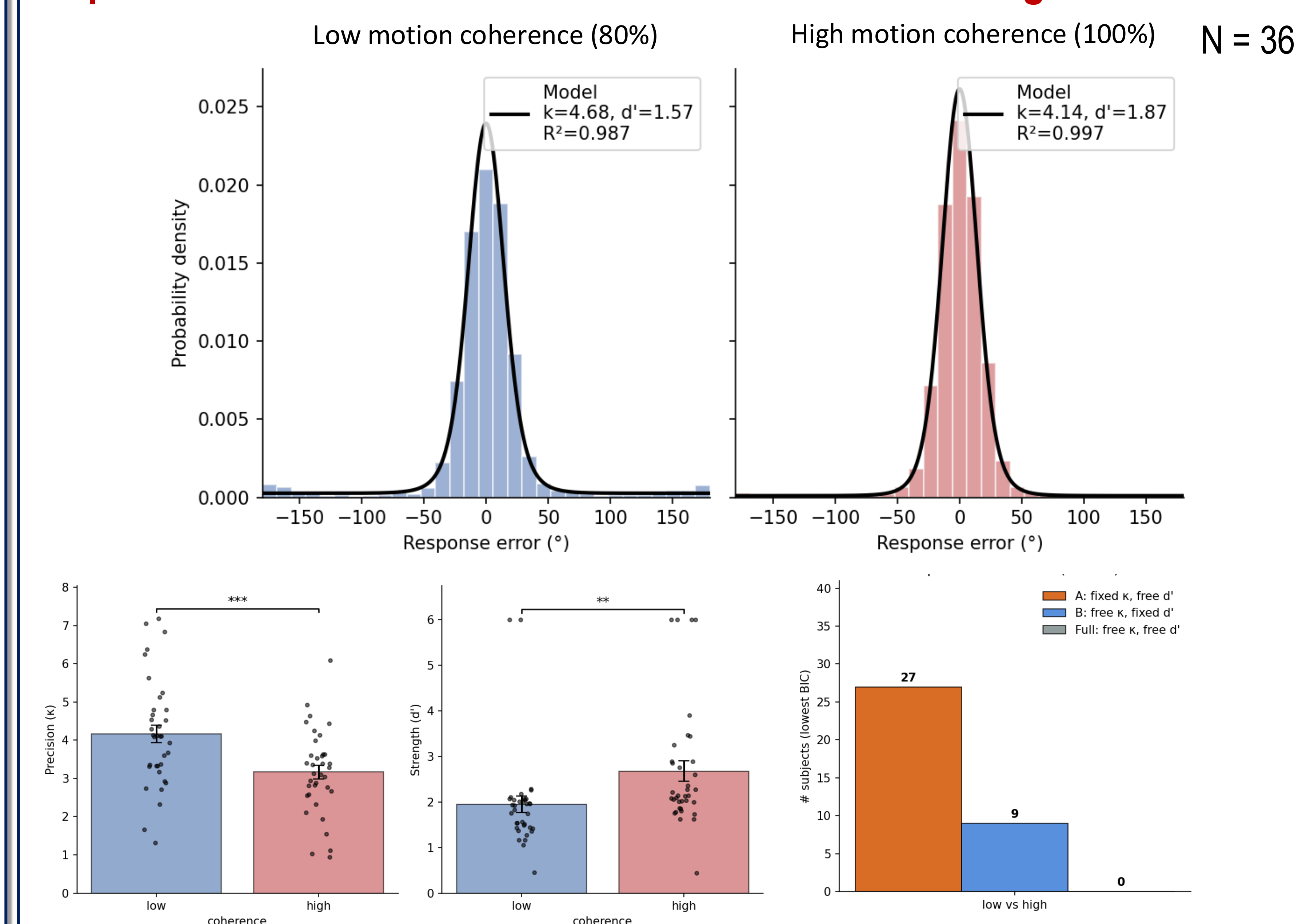
- Fit the TCC precision model to motion recall errors for each subject and condition
- Compare parameter estimates across experimental conditions
- Compare three models (full model, fixed- k , fixed- d') across subjects using BIC

Experiment 1: Motion variability will affect VWM precision



- TCCk model (full model) shows an excellent fit to aggregate data
- Increasing Gaussian noise (low to moderate) selectively decreased precision (k) with little impact on strength (d')
- Extremely large noise (SD 32 in deg) worsen both VWM precision and capacity
- Model comparison results show that fixed- d' model outperforms the others between low and middle noise conditions

Experiment 2: Motion coherence will affect VWM strength



- TCCk model (full model) shows an excellent fit to aggregate data
- Trade-off between precision and strength between low and high coherence conditions (coh 0.8 vs. 1.0)
- Model comparisons show that fixed- k model outperforms the other models, implying the variance of recalls is largely accounted for by changes in strength (d'), not precision (k)

Discussion and Conclusions

- Together, dissociable effects of Gaussian and white noise in motion direction on VWM precision and strength for global motion, suggesting d' likely to capture representational strength
- Consistent with the previous work⁴ on capacity and precision dissociation with distinct motion noise manipulations, suggesting a link between strength and capacity
- The two parameters are not fully independent; k and d' showed some shared explanatory variance, leading to trade-off in Exp 2 or the worse fit index of full model

References

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- Xie, W., Ma, T., Thakurdesai, S., Kim, I., & Zhang, W. (2024). Discrimination of mnemonic similarity is associated with short-term and long-term memory precision. *Memory & Cognition*. <https://doi.org/10.3758/s13421-024-01648-y>
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- Green, M., & Pratte, M. S. (2022). Local Motion Pooling is Continuous, Global Motion Perception is Discrete. *Journal of Experimental Psychology: Human Perception and Performance*. <https://doi.org/10.1037/xhp0000971.supp>

If you are interested in the mixture-model fits, please feel free to scan this



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<https://memory.ucr.edu/>

<https://inikim.github.io/ikim124.github.io/>